2021 Ph H2 Q1
Section: Our Dynamic Universe
Topic: Motion, Equations and Graphs

(a)(i)
To find the horizontal component of the velocity:

$$v_h = v\cos\theta = 16.0\cos(42^\circ) = 11.9\,\mathrm{m\,s^{-1}}$$

(a)(ii)
To find the vertical component of the velocity:

$$v_v = v\sin\theta = 16.0\sin(42^\circ) = 10.7\,\mathrm{m\,s^{-1}}$$

(b)
Vertical motion — time to reach maximum height:

$$v = u + at \Rightarrow 0 = 10.7 - 9.8t \Rightarrow t = \frac{10.7}{9.8} = 1.1\,\mathrm{s}$$

(c)
Horizontal distance travelled before landing:

$$s = v_h \times t = 11.9 \times 1.1 = 13.1\,\mathrm{m}$$

(Note: Question specifies the skier is in the air for 1.1 s.)

(d)
Greater
The skier lands at a lower level than takeoff, so gravitational potential energy has decreased.
By conservation of energy, this results in an increase in kinetic energy — hence, a **greater speed** at landing.

🧠 Revision Tips

- Always resolve vectors using sine (vertical) and cosine (horizontal).
- Vertical motion uses constant acceleration equations.
- Watch for whether total time of flight or just time to max height is needed.
- A decrease in height = gain in kinetic energy if no energy losses occur.